The κ-Source Framework

Theoretical Formulation, Mock Image Generation, and Hierarchical Extraction

Standalone High-Performance Rust Implementation | Software Version 0.1.0

Abstract

This document provides the formal theoretical definition and practical operating instructions for the **κ-Source Framework**. Under this formalism, any astronomical image is conceptualized as the union of κ-sources ($\kappa \in \{1, 2, ..., n\}$). We define a κ-source as a coherent cluster of κ constituent point sources within a spatial radius $R \leq R_{\max}$ whose individual flux is below standard detection thresholds ($F_i < 3$ RMS), but whose collective integrated flux satisfies $\sum F_i \geq 3$ RMS. In mock image synthesis, each point source subcomponent is explicitly convolved with the telescope Point Spread Function (PSF). We present high-performance Rust tools (`kappa_generate.bin` and `kappa_extract.bin`) to synthesize mock FITS images and perform hierarchical extraction from arbitrary FITS files.

1 Theoretical Concept of κ-Sources

1.1 The Fundamental Image Decomposition Principle

In conventional astronomical source detection (e.g., standard thresholding), images are decomposed solely into isolated point-like or extended components exceeding a local detection threshold (typically 3σ or 5σ). Faint sub-threshold components buried in noise are discarded.

Under the **κ-Source Formalism**, an astronomical sky model $I_{\text{sky}}(x, y)$ consists of constituent subcomponents that are convolved with the instrument Point Spread Function $\text{PSF}(x, y)$ and perturbed by background noise:

$$I_{\text{obs}}(x, y) = \left[I_{\text{sky}} \star \text{PSF}\right](x, y) + \mathcal{N}\left(0, \sigma_{\text{RMS}}^2\right)$$

where the sky is formed by the union of κ-sources:

$$I_{\text{sky}}(x, y) = \bigcup_{\kappa=1}^{n} \bigcup_{j=1}^{N_\kappa} S_{\kappa,j}(x, y)$$

1.2 Mathematical Definition of a κ-Source

A κ-Source $C_\kappa = \{s_1, s_2, ..., s_\kappa\}$ of multiplicity κ is formed by κ point sources satisfying two strict criteria:

1. **Spatial Coherence Constraint**: All κ constituent point sources reside within a maximum clustering radius $R_{\max}$ from their flux-weighted centroid:

$$\max_{i \in \{1..\kappa\}} \|\boldsymbol{x}_i - \boldsymbol{x}_{\text{cen}}\| \leq R_{\max}$$

2. **Collective Significance & Sub-threshold Condition**:
   - For $\kappa \geq 2$ (Multi-component κ-sources): Each constituent subcomponent i is individually below the point-source detection threshold:

$$\forall i \in \{1..\kappa\}, \quad F_i < 3.0 \times \sigma_{\text{RMS}}$$

while their collective integrated flux satisfies the detection threshold:

$$F_{\text{total}} = \sum_{i=1}^{\kappa} F_i \geq 3.0 \times \sigma_{\text{RMS}}$$

- For $\kappa = 1$ (1-Sources): The single isolated source meets detection on its own:

$$F_1 \geq 3.0 \times \sigma_{\text{RMS}}$$

Multiplicity κ	Individual Flux F_i	Collective Flux $\sum F_i$	Observational Classification
$\kappa = 1$ (1-source)	$F_1 \geq 3\sigma_{\text{RMS}}$	$F_1 \geq 3\sigma_{\text{RMS}}$	Directly detected point source
$\kappa = 2$ (2-source)	$F_i < 3\sigma_{\text{RMS}}$	$\sum F_i \geq 3\sigma_{\text{RMS}}$	Pair of faint subcomponents
$\kappa = 3$ (3-source)	$F_i < 3\sigma_{\text{RMS}}$	$\sum F_i \geq 3\sigma_{\text{RMS}}$	Triplet of faint subcomponents
$\kappa = n$ (n-source)	$F_i < 3\sigma_{\text{RMS}}$	$\sum F_i \geq 3\sigma_{\text{RMS}}$	Cluster of n faint subcomponents

1.3 Point Spread Function (PSF) Convolution

Each constituent subcomponent i is a point source with flux F_i at position (x_i, y_i):

$$I_{i,\,\text{sky}}(x, y) = F_i \delta(x - x_i, y - y_i)$$

Convolution with the telescope PSF produces the observed 2D intensity profile:

$$I_{i,\,\text{conv}}(x, y) = \left[I_{i,\,\text{sky}} \star \text{PSF}\right](x, y) = F_i \cdot \text{PSF}(x - x_i, y - y_i)$$

Supported PSF models:

- **2D Gaussian PSF**:

$$\text{PSF}(r) = \frac{1}{2\pi\sigma_{\text{PSF}}^2} \exp\left(-\frac{r^2}{2\sigma_{\text{PSF}}^2}\right), \quad \sigma_{\text{PSF}} = \frac{\text{FWHM}}{2\sqrt{2\ln 2}}$$

- **2D Moffat PSF** (Atmospheric Seeing):

$$\text{PSF}(r) = \frac{\beta - 1}{\pi\alpha^2}\left[1 + \frac{r^2}{\alpha^2}\right]^{-\beta}, \quad \alpha = \frac{\text{FWHM}}{2\sqrt{2^{\frac{1}{\beta}} - 1}}$$

1.4 Hierarchical Extraction Sequence

Extraction proceeds **hierarchically in order of increasing multiplicity**:

1. First, all **1-sources** ($\kappa = 1$) are identified and cataloged.
2. Next, all **2-sources** ($\kappa = 2$) are identified and cataloged.
3. Next, all **3-sources** ($\kappa = 3$), up to $\kappa = n$.

For every extracted κ-source, the following physical properties are measured:

- **Flux-weighted Centroid**:

$$\boldsymbol{x}_{\text{cen}} = (\bar{x}, \bar{y}) = \left(\frac{\sum_{i=1}^{\kappa} F_i x_i}{\sum_{i=1}^{\kappa} F_i}, \frac{\sum_{i=1}^{\kappa} F_i y_i}{\sum_{i=1}^{\kappa} F_i}\right)$$

- **Characteristic Spatial Extent (Radius)**:

$$R = \max_{i \in \{1..\kappa\}} \sqrt{(x_i - \bar{x})^2 + (y_i - \bar{y})^2} + \frac{\text{FWHM}}{2}$$

- **Signal-to-Noise Ratio (SNR)**:

$$\text{SNR} = \frac{F_{\text{total}}}{\sigma_{\text{RMS}}}$$

2 Multi-Extension FITS Architecture

All data generated and extracted are stored in standard multi-extension astronomical FITS files:

HDU	Name	Type	Contents
0	`PRIMARY`	Image HDU	2D Float array ($M \times M$) with metadata (`CONVOLV` , `PSF_TYPE` , `PSF_FWHM` , `NKAPPA` , `KAP_MAX` , `KAP_RAD` , `DET_SIG` , `BG_SIGMA`)
1	`SOURCES`	Binary Table	Point sources catalog (`ID` , `X` , `Y` , `FLUX` , `AMPLITUDE` , `SIGMA` , `FWHM` , `KAPPA_ID` , `KAPPA`)
2	`KAPPA_SRCS`	Binary Table	Extracted κ-sources catalog (`KAPPA_ID` , `KAPPA` , `CEN_X` , `CEN_Y` , `TOTAL_FLUX` , `RADIUS` , `SNR` , `N_MEMBERS`)

3 Practical Guide: Generating Mock FITS Images

The tool `kappa_generate.bin` synthesizes an $M \times M$ mock FITS image containing N κ-sources with multiplicity $\kappa \leq \kappa_{\max}$ and cluster radius $R \leq R_{\max}$ convolved with the telescope PSF on top of Gaussian background noise.

3.1 Command-Line Parameters

Option	Short	Default	Description
`--num-kappa`	`-N`	`50`	Total number of κ-sources to produce (N)
`--size`	`-M`	`4096`	Image grid dimension M ($M \times M$ pixels)
`--max-kappa`	`-k`	`5`	Maximum multiplicity bound ($1 \leq \kappa \leq \kappa_{\max}$)
`--max-radius`	`-r`	`25.0`	Maximum spatial cluster radius in pixels ($R_{\max}$)
`--psf`		`gaussian`	PSF profile model: `gaussian` or `moffat`
`--fwhm`		`10.0`	PSF Full Width at Half Maximum in pixels
`--moffat-beta`		`4.765`	Power-law index beta for Moffat PSF
`--detection-sigma`	`-s`	`3.0`	Collective flux threshold ($\sum F_i \geq S \times \text{RMS}$)
`--noise-sigma`		`1.0`	Background Gaussian noise standard deviation (RMS)
`--output`	`-o`	`mock_kappa_image.fits`	Output FITS file path

3.2 Usage Examples

```
# 1. Generate 50 kappa-sources convolved by Gaussian PSF on a 4096 x 4096 grid:
./kappa_generate.bin -N 50 -M 4096 -k 5 -r 25.0 --psf gaussian --fwhm 10.0 --output
mock_kappa.fits

# 2. Generate with Moffat atmospheric seeing PSF (FWHM = 8 px, beta = 4.765):
./kappa_generate.bin -N 60 -M 2048 -k 4 -r 20.0 --psf moffat --fwhm 8.0 -s 3.0 -o
moffat_sim.fits
```

4 Practical Guide: Extracting κ-Sources from FITS

The tool `kappa_extract.bin` ingests any arbitrary 2D FITS file (real or simulated) and extracts the κ-source hierarchy.

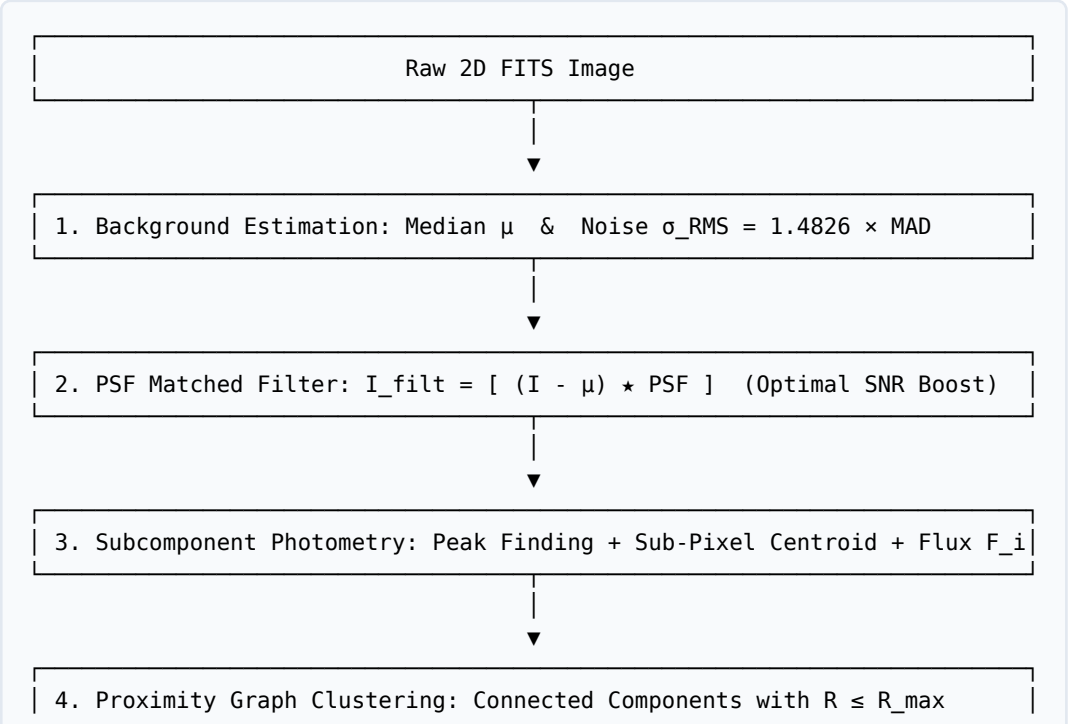
4.1 Five-Stage Extraction Strategy & Algorithm Pipeline

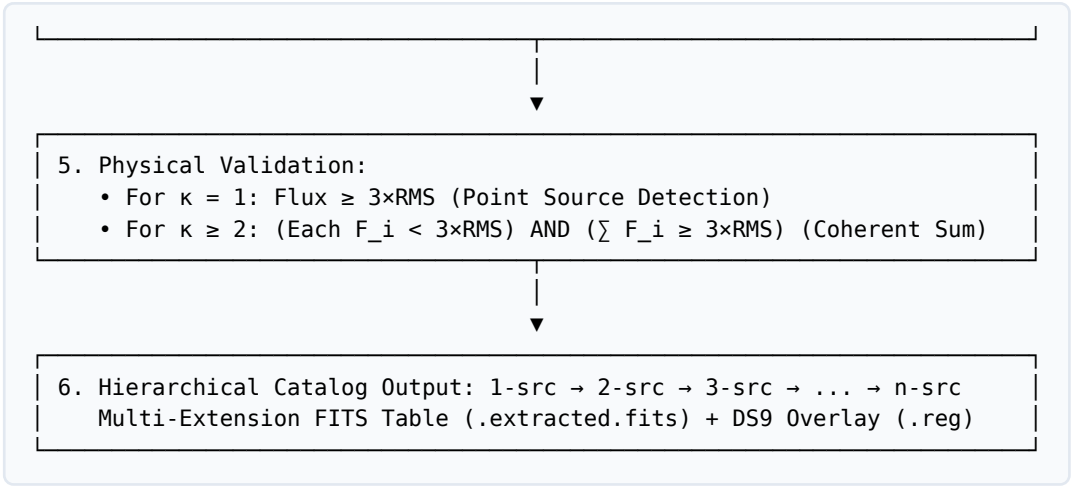
Extracting κ-sources from a noisy astronomical raster requires solving a dual problem: capturing faint sub-threshold peaks without triggering noise percolation, and testing whether spatial clusters reach statistical detection (≥ 3 RMS). The complete strategy comprises five sequential stages:

1. **Noise Background Characterization (σ_{RMS})**: Computes background median μ and robust dispersion via the Median Absolute Deviation:
$$\sigma_{\text{RMS}} = 1.4826 \times \text{median}(|I(x,y) - \mu|)$$
2. **PSF Matched Filtering (Spatial Optimal Filtering)**: Convolves the background-subtracted raster with the PSF kernel $I_{\text{filt}} = (I - \mu) \star \text{PSF}$. By the Matched Filter Theorem, this enhances real point-source peaks by a factor of $\approx \sqrt{2\pi\sigma_{\text{PSF}}^2} \approx 10\times$ while suppressing single-pixel high-frequency noise spikes.
3. **Subcomponent Peak Finding & Aperture Photometry**: Detects local maxima in the matched-filter map down to low candidate SNR (≥ 2.5). Sub-pixel centroids (x_i, y_i) are refined via 2D intensity moments, and total flux F_i is integrated using circular aperture photometry with analytic PSF aperture loss correction.
4. **Proximity Graph Clustering**: Constructs an adjacency graph where vertices are candidate peaks and edges connect neighbors with Euclidean separation $d(s_i, s_j) < 1.5R_{\max}$. Connected components of size κ are extracted, and their cluster radius $R_{\text{cluster}} \leq R_{\max}$ from the centroid is verified.
5. **Physical Validation & Hierarchical Sorting**: Tests each cluster against physical significance criteria:
   - **For $\kappa = 1$**: $F_1 \geq S_{\text{det}} \times \sigma_{\text{RMS}}$
   - **For $\kappa \geq 2$**: $(\forall i, F_i < S_{\text{det}} \times \sigma_{\text{RMS}})$ AND $\left(\sum_{i=1}^{\kappa} F_i \geq S_{\text{det}} \times \sigma_{\text{RMS}}\right)$

   Catalog records are sorted hierarchically starting with 1-sources, then 2-sources, 3-sources, …, n-sources.

4.2 Extraction Strategy Flowchart





4.3 Output Timestamped Catalog Files

To prevent accidental overwrites and keep session records organized, `kappa_extract.bin` automatically writes timestamped catalog products using the format `<fitsname>_<YYYYMMDD_HHMMSS>`:

- **FITS Catalog** (`<fitsname>_<timestamp>.extracted.fits`): Binary table HDU with full 32-bit float columns (`KAPPA_ID` , `KAPPA` , `CEN_X` , `CEN_Y` , `TOTAL_FLUX` , `MAX_AMP` , `RADIUS` , `SNR` , `N_MEMBERS`).
- **ASCII Table** (`<fitsname>_<timestamp>.extracted.cat`): Plain text table with metadata header cards and aligned columns.
- **CSV Catalog** (`<fitsname>_<timestamp>.extracted.csv`): Comma-separated format for direct loading into Pandas, Astropy, or Excel.
- **DS9 Region File** (`<fitsname>_<timestamp>.extracted.reg`): Color-coded region overlay.

4.4 Command-Line Parameters

Option	Short	Default	Description
`<INPUT>`		**(required)**	Path to input FITS image (e.g. `test.fits`)
`--max-kappa`	`-k`	`0` **(all)**	Maximum multiplicity upper limit (e.g. `-k 3` restricts to $\kappa \leq 3$)
`--detection-sigma`	`-s`	`3.0`	Collective flux detection threshold ($\geq S \times$ Beam RMS)
`--search-radius`	`-r`	`25.0`	Search radius R_{search} in pixels from centroid
`--fwhm`		`10.0`	Estimated PSF FWHM in pixels (set 0 or use `--psf-auto` for auto mode)
`--psf-auto`		`false`	Automatically measure PSF FWHM from bright isolated point sources
`--min-sub-snr`		`1.2`	Minimum candidate peak SNR for subcomponents in matched filter
`--seed-snr`		`2.2`	Candidate cluster seed threshold on smoothed map (alias: `--peak-snr`)
`--output`	`-o`	`<auto>`	Custom output FITS catalog path (defaults to timestamped name)

4.5 Usage Examples

```
# 1. Automatic extraction measuring PSF FWHM directly from the image:
./kappa_extract.bin test.fits --psf-auto -s 3.0 -r 25.0

# 2. Extract with known FWHM and 5xRMS detection threshold:
./kappa_extract.bin test.fits --fwhm 10.0 -s 5.0 -r 30.0
```

5 Visualization with SAOImage DS9 & Python

5.1 DS9 Color-Coded Overlay

Both `kappa_generate.bin` and `kappa_extract.bin` automatically export companion `.reg` files color-coded by multiplicity hierarchy:

- 🟢 **Green circles**: 1-Sources ($\kappa = 1$)
- 🟡 **Yellow circles**: 2-Sources ($\kappa = 2$)
- 🟠 **Orange circles**: 3-Sources ($\kappa = 3$)
- 🔴 **Red circles**: $\kappa \geq 4$-Sources
- ⚪ **Dashed white circles**: Constituent subcomponent peaks

Open the image and region overlay with one command:

```
ds9 test.fits -regions test.extracted.reg -scale zscale -zoom to fit
```

5.2 Inspecting Catalogs in Python (Astropy)

```
from astropy.io import fits

with fits.open("test.extracted.fits") as hdul:
    hdul.info()

    # 1. Access extracted kappa-sources catalog
    ktable = hdul["KAPPA_SRCS"].data
    print(f"Total kappa-sources extracted: {len(ktable)}")
    for row in ktable[:5]:
        print(f"ID={row['KAPPA_ID']}: kappa={row['KAPPA']}, Pos=({row['CEN_X']:.1f},
{row['CEN_Y']:.1f}), Flux={row['TOTAL_FLUX']:.3f}, SNR={row['SNR']:.1f}")
```